

A multi-expedition soil microbiome and geochemistry dataset from the Rub’ al-Khali desert

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Abstract

Background. The Rub’ al-Khali (Empty Quarter) is the largest contiguous sand desert on Earth and an understudied extreme environment whose microbial communities and biogeochemistry remain poorly characterized. Integrated, cross-domain data resources for such ecosystems are scarce: microbial sequencing, soil chemistry, and environmental observations are typically deposited in disconnected repositories with incompatible schemas, which limits reuse and cross-domain analysis. **Findings.** We present a coordinated multi-expedition dataset comprising 16S rRNA gene amplicon sequencing profiles from 2,535 soil samples collected at 70 sampling sites along a 1,043 km transect across the Rub’ al-Khali during five field expeditions (2023–2025); X-ray fluorescence (XRF) measurements of 91 chemical analytes from 249 measurement sessions; GPS-tagged environmental metadata (temperature, pressure, humidity) recorded at every site visit; and historical monthly and annual climate records from the Open-Meteo database. Taxonomic classifications from multiple primer and classifier workflows are harmonized against NCBI Taxonomy and the Genome Taxonomy Database (GTDB), yielding 1,401,008 taxon-sample abundance observations. All data are integrated into a formal knowledge graph that encodes the full provenance chain from field sampling to digital data products; the knowledge graph comprises approximately 37 million asserted RDF triples. **Conclusions.** The dataset is published as raw sequencing data in the European Nucleotide Archive (umbrella project PRJEB104209, amplicon project PRJEB106069), as versioned OWL/RDF modules distributed from the project website and the Zenodo archive, and via a public SPARQL endpoint at <https://rubalkhali.science/sparql>. All processing code is released under a free software licence. A companion paper describes the formal knowledge graph and the Semantic Web engineering that supports cross-domain querying.

Keywords. desert microbiome; metagenomics; geochemistry; X-ray fluorescence; 16S rRNA; knowledge graph; Rub’ al-Khali; Saudi Arabia; FAIR data

1 Data description

The Rub’ al-Khali is the largest contiguous sand desert on Earth, covering approximately 650,000 km² across the southern Arabian Peninsula [17]. Despite extreme aridity and surface temperatures that frequently exceed 50°C, the desert harbours microbial communities whose composition, diversity, and ecological drivers remain poorly characterized [16]. Predicting ecosystem responses to climate change in such environments and informing conservation and land-use policy require coordinated, multi-variable datasets that capture microbial, chemical, and environmental information at matching spatial and temporal resolutions [11]. Current resources fall short: microbial sequencing data are typically deposited in sequence archives, soil chemistry data in laboratory spreadsheets, and environmental observations in climate databases, with no common identifiers or data model linking them. This fragmentation blocks the cross-domain analyses that extreme-environment microbial ecology requires.

This Data Note presents a coordinated dataset that addresses the gap for the Rub’ al-Khali. The dataset is the product of five field expeditions conducted across seasons between early 2023 and late 2025, covering 70 sampling sites along a 1,043 km transect from the western

escarpment of the desert to the Saudi Arabia–Oman border. At each site we collected bulk surface, bulk subsurface (“deep”), and rhizosphere soil samples in three biological replicates, yielding 2,535 soil samples in total. We generated 16S rRNA amplicon sequencing profiles for all samples on Illumina MiSeq, processed them with a standardized workflow (nf-core/Ampliseq + DADA2), and harmonized the resulting taxonomic classifications across primer and classifier combinations by aligning NCBI Taxonomy, the Genome Taxonomy Database (GTDB) [19], and the iNaturalist taxonomy into a single reference. For a subset of the sites we generated X-ray fluorescence measurements of 91 soil analytes in 249 independent measurement sessions. At every site visit we recorded temperature, atmospheric pressure, and relative humidity, and we additionally retrieved historical climate data from the Open-Meteo database [18] to provide 12,936 monthly and 582 annual climate records.

All data are integrated into a formal, semantic knowledge graph that links physical samples to their digital data products through an explicit provenance chain. We used the Semantic-science Integrated Ontology (SIO) [6] as the upper-level framework, grounding chemical entities in the Chemical Entities of Biological Interest (ChEBI) ontology [5], environmental contexts in the Environment Ontology (ENVO) [3], and taxonomic identifiers in NCBI Taxonomy [22]. The knowledge graph comprises approximately 37 million asserted RDF triples served through a public SPARQL 1.1 endpoint. This Data Note describes the data acquisition, processing, validation, and access methods; the semantic modelling, ontology design patterns, materialization strategy, and reasoning evaluation are described in a companion paper targeting the Semantic Web community [9].

2 Methods

2.1 Study area and sampling design

We conducted five sampling expeditions across the Rub’ al-Khali along a transect from the western escarpment (19.15°N, 45.17°E) to the eastern border of Saudi Arabia with Oman (20.71°N, 54.73°E), spanning 1,043 km (Figure 1). The transect encompasses 60 primary sampling sites at predetermined geographic coordinates, together with 10 sites of special interest (water wells, a hot spring, an oil spill site), for a total of 70 sampling sites.

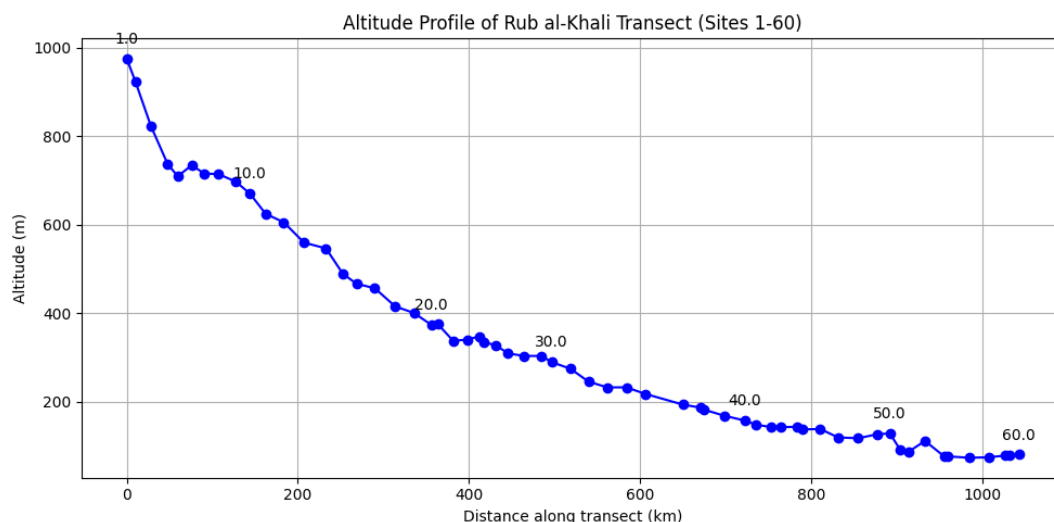


Figure 1: Altitude profile along the sampling transect across the Rub’ al-Khali, from the western escarpment to the eastern border with Oman.

To capture temporal variability, we conducted expeditions across different seasons: late

winter 2023 (Trip 1, 64 sites visited); summer 2023 (Trip 2, truncated to 8 sites due to extreme temperatures); winter 2024 (Trip 3, 65 sites visited); summer 2024 (Trip 4, 60 sites visited); and autumn 2025 (Trip 5, 62 sites visited). Table 1 summarizes the observational coverage. At each site visit we recorded temperature, atmospheric pressure, and relative humidity with handheld field instruments, and during one expedition we additionally collected XRF measurements to characterize the geochemical composition of the soils.

Table 1: Summary of expedition coverage and environmental measurements collected at each site visit.

Expedition	Season	Sites visited	Environmental measurements
Trip 1	Late winter 2023	1–60, 4 special	Temp., press., hum.
Trip 2	Summer 2023	1–8	Temp., press.
Trip 3	Winter 2024	1–60, 5 special	Temp., press., hum.
Trip 4	Summer 2024	1–60	Temp., hum.
Trip 5	Autumn 2025	1–60, 2 special	Temp., press., hum.

At each site we collected samples from three distinct soil compartments: bulk surface soil (0 cm depth), bulk subsurface soil (5–10 cm depth, hereafter “deep”), and subsurface soil near plant roots (3–10 cm depth, hereafter “rhizosphere”). We obtained three biological replicates per compartment by placing soil into sterile tubes, either by scooping soil directly into the tube or using sterile plastic spoons. We preserved samples at -20°C in the field using an Engel MR040F portable freezer; at the end of each expedition we transferred samples to a -80°C freezer in the laboratory until processing.

We included positive and negative controls for each sampling campaign to control for contamination artefacts. Positive controls consisted of replicate tubes of Zymo Mock Community Standard (ZymoBIOMICS, cat. #D6305). Negative controls consisted of sterilized soil treated with UV light for four hours. Controls accompanied every batch through DNA extraction, library preparation, and sequencing.

2.2 DNA extraction and sequencing

We utilized 0.9–1.2 g of soil for DNA extraction using either the PowerLyzer PowerSoil Kit (QIAGEN, cat. #12855-50) or the DNeasy PowerSoil Pro Kit (QIAGEN, cat. #47014). We homogenized all samples with the PowerLyzer 24 Homogenizer and completed extraction either manually or via the automated QIAcube Connect system. We quantified eluted DNA using the Qubit HS dsDNA kit (Thermo Fisher Scientific, cat. #Q32854).

We amplified bacterial DNA with the Bakt_341F [13] and 806R [1] primer set, targeting the V3–V4 region of the 16S rRNA gene. We performed amplification using the Multiplex PCR kit (QIAGEN) and sequenced the resulting amplicon libraries on an Illumina MiSeq platform in a 2×250 paired-end configuration. All library preparation and sequencing was performed in the principal investigator’s laboratory at KAUST.

2.3 Sequencing data processing

We processed raw amplicon reads with nf-core/Ampliseq (v2.14.0) [10,23] and QIIME2 (v2024.10) [2]. We applied the DADA2 [4] algorithm for error correction, chimera removal, and denoising, producing amplicon sequence variants (ASVs) at single-nucleotide resolution. We identified taxa using the Genome Taxonomy Database (GTDB) [19]. We retained ASVs that were present in at least three samples with a total abundance greater than 50 reads. We identified and removed contaminant reads using the QIIME2 *decontam* module based on our negative control samples, applying a prevalence threshold of 0.52.

2.4 Taxonomy harmonization

To unify taxonomic classifications from different reference databases, we developed an algorithm that maps and aligns the NCBI Taxonomy (O_{ncbi}), the Genome Taxonomy Database (O_{gtdb}), and the iNaturalist taxonomy (O_{inat}). The alignment does not imply any biological commitment but integrates three reference taxonomies for the purposes of cross-expedition queries, resolving nomenclature divergence (for example, lineage-specific alphanumeric suffixes in GTDB) and structural rank conflicts into a consolidated taxonomic module O_{final} .

Let $O = (C, R, E)$ represent a taxonomy comprising a set of classes C , a set of taxonomic ranks R , and a set of subclass axioms E . We define a lexical normalization function $f_{norm} : C \rightarrow \Sigma^*$ that maps a class $c \in C$ to a canonical string by converting the primary label to lowercase, removing rank-specific prefixes (for example $d_$), and stripping alphanumeric lineage suffixes. Let $rank : C \rightarrow R$ denote a function that retrieves the normalized taxonomic rank of a class, and let $parent : C \rightarrow C$ denote a function that retrieves the direct superclass of c .

The workflow initializes with a lexical alignment phase that generates a candidate seed set

$$S_{lex} = \{ (c_{src}, c_{tgt}) \in C_{ncbi} \times C_{gtdb} \mid f_{norm}(c_{src}) = f_{norm}(c_{tgt}) \}.$$

Lexical matching between NCBI and GTDB produced 27,487 seed pairs. We subsequently partitioned S_{lex} through a structural verification step that evaluates rank compatibility and topological support. A seed pair $(c_{src}, c_{tgt}) \in S_{lex}$ is classified as a strong anchor A_{strong} if $rank(c_{src}) = rank(c_{tgt})$ and $f_{norm}(parent(c_{src})) = f_{norm}(parent(c_{tgt}))$. Pairs whose ranks are compatible but whose parent labels diverge are classified as weak anchors A_{weak} ($|A_{weak}| = 4,347$), which signify potential nomenclature reclassifications.

We resolve weak anchors using a bottom-up inference strategy based on the biological heuristic that identical species-level designations refer to the homologous entity regardless of divergent higher-rank classifications. For each $(c_{src}, c_{tgt}) \in A_{weak}$, we generate a structural hint $h = (parent(c_{src}), parent(c_{tgt}))$. We aggregate the hints and apply an equivalence axiom to higher-level taxa if the vote count for a structural hint exceeds a predefined threshold, yielding 1,514 parent hints. We then graft the iNaturalist taxonomy into the merged NCBI+GTDB module by matching iNaturalist classes through the same normalization function, appending equivalence axioms for singular matches and disambiguating multi-match cases by rank annotation. The full alignment procedure is given in Algorithm 1.

The alignment process successfully produced 27,487 verified anchors bridging NCBI and GTDB. Following the primary integration, we anchored the iNaturalist taxonomy (O_{inat}) into the merged NCBI+GTDB module. The merged module contains 5,878 unique taxa observed across the five expeditions, mapped onto the reference taxonomies while preserving the original classifier assertions so that downstream consumers can re-use the mapping or substitute their own.

2.5 X-ray fluorescence analysis

We measured the elemental composition of soil samples with a portable energy-dispersive X-ray fluorescence (ED-XRF) spectrometer. Each measurement session generated concentration values for 91 chemical analytes, including major oxides and trace elements, expressed as mass percentages. The instrument reports a ‘‘Light Elements’’ (LE) pseudo-analyte that captures the summed contribution of elements below the detection limit of the ED-XRF method (primarily H, Li, Be, B, C, N, O, F, Na, and Mg); we carry this through as a separate pseudo-class in the knowledge graph so that it does not bias the per-element concentration totals. We applied detector-sensitivity corrections to the identified elements and scaled the corrected concentrations so that the final composition totals 100% ($\sum \text{final concentrations} + \text{LE} = 100\%$). We mapped each analyte to the Chemical Entities of Biological Interest (ChEBI) ontology [5] and to the PubChem database [15]. Of the 93 monitored parameters, 80 analytes mapped directly onto

Algorithm 1 Taxonomy alignment and integration workflow.

```
1: Input:  $O_{ncbi}$ ,  $O_{gtdb}$ ,  $O_{inat}$ 
2: Output:  $O_{final}$ 
3:  $S_{lex} \leftarrow \emptyset$ ;  $A_{strong} \leftarrow \emptyset$ ;  $A_{weak} \leftarrow \emptyset$ 
4: // Phase 1: lexical anchoring
5: for  $(c_{src}, c_{tgt}) \in C_{ncbi} \times C_{gtdb}$  do
6:   if  $f_{norm}(c_{src}) = f_{norm}(c_{tgt})$  then
7:      $S_{lex} \leftarrow S_{lex} \cup \{(c_{src}, c_{tgt})\}$ 
8:   end if
9: end for
10: // Phase 2: structural verification
11: for  $(c_{src}, c_{tgt}) \in S_{lex}$  do
12:   if  $f_{norm}(\text{parent}(c_{src})) = f_{norm}(\text{parent}(c_{tgt}))$  then
13:      $A_{strong} \leftarrow A_{strong} \cup \{(c_{src}, c_{tgt})\}$ 
14:   else
15:      $A_{weak} \leftarrow A_{weak} \cup \{(c_{src}, c_{tgt})\}$ 
16:   end if
17: end for
18: // Phase 3: bottom-up resolution
19:  $H \leftarrow \text{AGGREGATEPARENTHINTS}(A_{weak})$ 
20:  $O_{merged} \leftarrow O_{ncbi} \cup O_{gtdb}$ 
21: for  $(c_{src}, c_{tgt}) \in A_{strong} \cup H$  do
22:    $O_{merged} \leftarrow O_{merged} \cup \{\text{EQUIVALENTCLASSES}(c_{src}, c_{tgt})\}$ 
23: end for
24:  $O_{merged} \leftarrow \text{DISAMBIGUATEHOMONYMS}(O_{merged})$ 
25: // Phase 4: iNaturalist grafting
26:  $O_{final} \leftarrow O_{merged} \cup O_{inat}$ 
27: for  $c_{inat} \in C_{inat}$  do
28:    $M \leftarrow \{c \in C_{merged} \mid f_{norm}(c) = f_{norm}(c_{inat})\}$ 
29:   if  $|M| = 1$  then
30:      $O_{final} \leftarrow O_{final} \cup \{\text{EQUIVALENTCLASSES}(c_{inat}, m \in M)\}$ 
31:   else if  $|M| > 1$  then
32:      $m^* \leftarrow \text{DISAMBIGUATEBYRANK}(M, \text{rank}(c_{inat}))$ 
33:      $O_{final} \leftarrow O_{final} \cup \{\text{EQUIVALENTCLASSES}(c_{inat}, m^*)\}$ 
34:   end if
35: end for
36: return  $O_{final}$ 
```

ChEBI classes and 92 parameters aligned to PubChem Compound Identifiers. In total, the XRF data comprise 6,674 individual measurement values across 249 measurement sessions.

2.6 Environmental and climate data

We retrieved historical climate data from the Open-Meteo database [18] for each sampling site, yielding 12,936 monthly and 582 annual records encompassing temperature, total precipitation, rainfall, and relative humidity. Field researchers manually classified each site according to its biome (for example desert biome, hot desert biome) and specific environmental features (for example sand dune, saline pan, gravel pavement, desert oasis). We subsequently mapped each classification to a class in the Environment Ontology (ENVO) [3], so that every site is linked to a controlled biome IRI and to one or more environmental-feature IRIs.

2.7 Data integration and structural validation

We integrated all data into a formal knowledge graph by generating Resource Description Framework (RDF) assertions for every sample, process, measurement, and environmental observation. The knowledge graph uses the SemanticScience Integrated Ontology (SIO) [6] as the upper-level vocabulary and models processes, qualities, and quantities as subclasses of the corresponding SIO classes. Domain-specific classes are defined as subclasses of SIO entities; chemical entities are grounded in ChEBI, environmental contexts in ENVO, and taxonomic identifiers in NCBI Taxonomy. The resulting knowledge graph is released as nine modular OWL/RDF files (see Section 3). The modelling details, formal design patterns, and reasoning workflow are described in the companion paper [9].

To ensure structural integrity during ingest, we defined Shape Expressions (ShEx) [20] shapes for every core entity type: sampling sites, soil samples, environmental measurements, DNA extracts, amplicon libraries, sequencing datasets, quality control records, and taxonomy observations. Each shape specifies the mandatory properties, their cardinality, and admissible datatypes (for example, that every sampling site must declare a label and at least one ENVO biome link, and that every DNA extract must carry a quantitative concentration measurement). The validator is implemented in the Apache Jena ShEx library (version 4.10.0) and is invoked from the data-generation workflow: every time a new ABox module is produced it is checked against its shape, and the workflow aborts if any instance violates the schema. This mechanism acts as a regression test so that modifications to the extract-transform-load scripts cannot silently introduce structural errors. All eight shape-annotated ABox modules passed validation without constraint violations.

We additionally verified the logical consistency of the knowledge graph using the ELK reasoner [14], an implementation of a consequence-based reasoning procedure for the OWL 2 EL profile. ELK confirmed that the ontology is consistent (zero unsatisfiable classes) both before and after the OWL-Horst materialization step performed by Virtuoso during deployment. The details of the reasoning evaluation, including the materialization benchmarks, are reported in the companion paper [9].

3 Data records

All raw sequencing data generated in this study are deposited at the European Nucleotide Archive (ENA). Project and run accessions are listed in Table 2. Each FASTQ dataset in the knowledge graph is cross-referenced to its ENA accession via the `rdfs:seeAlso` predicate.

We distribute the integrated knowledge graph as nine modular OWL/RDF files listed in Table 3. Each file is self-contained: consumers interested in a specific data type (for example only the sampling sites and their biome annotations, or only the XRF data) can download the single file they need without pulling the full graph. All modules are available as OWL/XML

Table 2: ENA project accessions.

Accession	Scope	Link
PRJEB104209	Umbrella project	https://www.ebi.ac.uk/ena/browser/view/PRJEB104209
PRJEB106069	16S rRNA amplicons	https://www.ebi.ac.uk/ena/browser/view/PRJEB106069

from the project download page (<https://rubalkhali.science/data>) and are archived with a persistent identifier in the Zenodo deposit [7] and as a curated GigaDB entry [8]. The SPARQL 1.1 endpoint at <https://rubalkhali.science/sparql> serves the same data for live querying, and a web portal at <https://rubalkhali.science/> provides an interactive map and per-site dashboards for users who do not wish to write SPARQL.

Table 3: Distributed knowledge graph modules.

File	Contents
<code>rubalkhali.owl</code>	Base ontology (TBox); 302 domain classes grounded in SIO, ChEBI, ENVO, PATO, UO.
<code>rubalkhali_sites.owl</code>	70 sampling sites with coordinates and ENVO biome annotations.
<code>rubalkhali_samples.owl</code>	2,535 soil samples with compartment type (surface, deep, rhizosphere) and replicate identifier.
<code>rubalkhali_measurements.owl</code>	Environmental measurements (temperature, pressure, humidity) together with 12,936 monthly and 582 annual climate records.
<code>rubalkhali_xrf.owl</code>	6,674 XRF measurement values across 91 analytes from 249 measurement sessions.
<code>rubalkhali_dna.owl</code>	2,410 DNA extracts with concentration measurements, extraction agents and kits, and protocol links.
<code>rubalkhali_sra.owl</code>	1,242 sequencing run records linked to ENA accessions via <code>rdfs:seeAlso</code> .
<code>rubalkhali_qc.owl</code>	Sequencing quality metrics (GC content, duplicate rate, sequence counts) for every run.
<code>rubalkhali_taxonomy_abox.owl</code>	1,401,008 taxon-sample abundance observations with absolute counts and relative proportions.

4 Technical validation

We validated the dataset using four complementary approaches: control samples processed alongside every sequencing batch; structural validation of every RDF module against formal Shape Expressions schemas; logical consistency checking of the integrated knowledge graph with a description-logic reasoner; and competency question evaluation via SPARQL queries that emulate the intended query workload.

Control samples. Positive and negative controls were processed through the full extraction, amplification, and sequencing workflow in every expedition batch. Zymo Mock Community standards yielded the expected taxonomic composition within the error bounds reported by the manufacturer, confirming the amplification and classification pipeline. UV-sterilized negative controls carried no reproducible off-target sequences; contaminant ASVs flagged by the QIIME2 *decontam* module (prevalence threshold 0.52) were removed prior to downstream integration.

Structural validation. We integrated the Apache Jena ShEx validator into the automated data-generation workflow so that every ABox module is tested against its shape immediately after generation and before loading into the triple store. The validator checks that mandatory properties are present, that cardinalities are respected, and that datatypes are consistent with the expected domain. The workflow aborts with an explicit error message if any instance violates a shape constraint, for example if a sampling site lacks GPS coordinates or a DNA extract lacks a concentration value. All eight shape-annotated ABox modules (`rubalkhali_sites.owl`, `rubalkhali_samples.owl`, `rubalkhali_measurements.owl`, `rubalkhali_dna.owl`, `rubalkhali_xrf.owl`, `rubalkhali_sra.owl`, `rubalkhali_qc.owl`, and `rubalkhali_taxonomy_abox.owl`) passed ShEx validation without constraint violations.

Logical consistency. We verified the logical consistency of the full knowledge graph (base ontology, all ABox modules, and the taxonomy reference module) with the ELK reasoner [14]. ELK confirmed that the ontology is consistent—no class is reduced to the empty set and no individual belongs to disjoint classes—both before and after OWL-Horst forward-chaining materialization has been applied by the deployment engine. We chose ELK over more expressive reasoners such as HermiT [12] because the 37-million-triple ABox renders full OWL 2 DL reasoning computationally infeasible on commodity hardware. The consequence-based calculus implemented by ELK produces class-hierarchy results under the OWL 2 EL profile in under two seconds on the asserted graph, and the post-materialization run confirms that the additional 5.7 million type assertions do not introduce disjointness violations. Full reasoning benchmarks are reported in the companion paper [9].

Competency questions. We defined nine competency questions expressed as SPARQL queries with known expected answers, covering (i) per-analyte XRF retrieval, (ii) site listings with GPS coordinates, (iii) biome assignments via OWL restrictions, (iv) Light Elements percentages, (v) DNA extraction records, (vi) provenance tracing from soil sample through library preparation and sequencing to ENA run accession, (vii) taxonomic composition per sequencing run, (viii) cross-domain queries joining environmental variables with taxon abundances, and (ix) per-sample quality control metrics. All nine queries return the expected result sets against the deployed endpoint. The queries are executed as part of continuous integration so that any regression in the data or in the ontology surfaces as a test failure; full queries and latency benchmarks appear in the companion paper [9].

5 Re-use potential

The dataset supports several re-use scenarios. At the smallest scale, individual investigators can retrieve the taxonomic composition and accompanying environmental metadata for a specific site or trip by downloading the relevant ABox modules or by issuing SPARQL queries against the endpoint. At the largest scale, the full knowledge graph supports cross-domain queries that would be difficult or impossible to express against a conventional relational store. For example, researchers interested in microbial responses to thermal stress can retrieve the relative abundance of particular taxa at every site where the recorded field temperature exceeds a threshold, with the SPARQL query traversing soil sample, DNA extract, amplicon library, sequencing run, FASTQ dataset, and relative-abundance observation in a single go. Researchers interested in geochemistry can ask for XRF concentration patterns grouped by ENVO biome, joining the site’s biome assignment with the per-sample analyte values.

The modular OWL architecture also makes it straightforward to extend the knowledge graph with new data types. Researchers can introduce new classes as subclasses of existing SIO entities and generate corresponding ABox modules; the existing ShEx shapes can be extended to cover the new classes, and the re-materialization workflow in Virtuoso rebuilds the inferred triples

automatically. Candidate extensions include metabolomics, remote sensing products, biogeography observations, and additional seasonal field trips. Because the identifiers are resolvable and the ontology is grounded in widely-used upper-level vocabularies, third parties can also link their own data into the knowledge graph by asserting equivalence or relationship axioms against our IRIs.

Users who do not wish to write SPARQL can access the data through the web portal at <https://rubalkhali.science/>, which provides an interactive map of sampling sites and per-site dashboards with environmental measurements, geochemistry, and taxonomic profiles. The portal also hosts a SPARQL editor pre-populated with the nine competency questions as click-to-load examples, together with an autocomplete-aware query editor (YASGUI) [21]. The same editor is used internally as a regression test for the live endpoint.

6 Availability of supporting data and materials

Sequencing reads. Raw 16S rRNA gene amplicon reads are deposited at the European Nucleotide Archive under the umbrella project PRJEB104209 (<https://www.ebi.ac.uk/ena/browser/view/PRJEB104209>) and the amplicon sub-project PRJEB106069 (<https://www.ebi.ac.uk/ena/browser/view/PRJEB106069>).

Knowledge graph modules. All OWL/RDF modules listed in Table 3, together with the ShEx shape definitions, the SPARQL competency questions, the schema documentation, and the ontology design pattern documentation, are archived on GigaDB [8] and on Zenodo [7]. The same files are served live from the project download page at <https://rubalkhali.science/data>. A public SPARQL 1.1 endpoint is available at <https://rubalkhali.science/sparql>.

Source code. All source code for data processing, ontology engineering, RDF generation, ShEx validation, the SPARQL competency questions, the web portal (React + FastAPI), and the Docker Compose deployment is publicly available on GitHub at <https://github.com/bio-ontology-research-group/empty-quarter>. The repository includes Groovy scripts that use the OWL API 5 for OWL/RDF generation, Python scripts for data processing and statistical analysis, and the deployment scripts that materialize the graph in Virtuoso.

License. The dataset, knowledge graph, ontology, ShEx shapes, and source code are released under the Creative Commons Attribution 4.0 International licence (CC BY 4.0, <https://creativecommons.org/licenses/by/4.0/>).

Declarations

List of abbreviations. ABox, Assertion Box; ASV, amplicon sequence variant; CC BY, Creative Commons Attribution; ChEBI, Chemical Entities of Biological Interest; DL, Description Logic; ED-XRF, energy-dispersive X-ray fluorescence; ELK, EL reasoner; ENA, European Nucleotide Archive; ENVO, Environment Ontology; ETL, extract–transform–load; FAIR, Findable, Accessible, Interoperable, Reusable; GTDB, Genome Taxonomy Database; IRI, Internationalized Resource Identifier; LE, Light Elements; NCBI, National Center for Biotechnology Information; OWL, Web Ontology Language; PATO, Phenotype and Trait Ontology; QC, quality control; RDF, Resource Description Framework; SHACL, Shapes Constraint Language; ShEx, Shape Expressions; SIO, Semanticscience Integrated Ontology; SPARQL, SPARQL Protocol and RDF Query Language; TBox, Terminological Box; UO, Units Ontology; XRF, X-ray fluorescence.

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